

Qty	Ref	Part
✓2	C1,5	0.1uF radial 0.1"LS.
1	C2	2.2uF 16v 0.2"LS tantalum. "+" end to match "+" on PCB!
✓4	C3,4,6,7	0.047uF 50v axial 0.3"LS.
✓3	D1,2,3	1N5818 Schottky diode. Banded end to match PCB!
✓2	D4,5	red T1 LED. Short wire in hole closest to 7-seg LEDs!
3	LED1,2,3	3-digit 7-seg red LED display ebay#404676137923 3361AS.
1	P1	6-pin Molex KK for CH340 USB-serial adapter. Bend pins 90 deg.
	0R	6-pin rt.angle header for Sparkfun USB-serial adapter, no pin 2. Only install ONE to suit your USB adapter.
2	P2	AAA cell clips, Keystone #82.
3	P3,4,5	3-pin header 0.1"LS.
3	P3A,4A,5A	jumpers for 3-pin headers
1	Q1	FJN4303 PNP transistor w.22k resistors. Flat side to match PCB!
✓1	R1	1meg 5% 1/4w resistor.
✓1	R2	39K 5% 1/4w resistor.
2	R3,4	4.7Kx9 bussed 10-pin SIP10.
✓1	R5	4.7K 5% 1/4w resistor(have 100 1/8w)
16	S0-9,R,P,M	+,+,-,* tactile pushbutton switch 0.25"LS.
1	SP1	piezo speaker 0.15"LS 0.7"dia.
1	SW	slide switch 0.08"LS, ebay#176298432688.
1	U1	OKI M80C85A microprocessor.
1	U1A	40-pin socket strip. Cut into two 20-pin strips for U1.
1	U2	74HC573 octal latch.
1	U3	39SF040 512K flash ROM (use 27C256, or ebay#389352133150)
1	U3A	40-pin socket strip. Cut into two 14-pin strips for U3.
1	U4	CY7C199 32K RAM.
1	U5	74HC251 decoder.
1	U6	OKI M82C55A 3-I/O ports (sent i82C55), or ebay#178069030834.
1	U6A	40-pin socket strip. Cut into two 20-pin strips for U6.
1	U7	74LS145 1-of-10 decoder.
1	U8	4093 quad 2-input NAND Schmitt trigger.
1	Y1	4MHz resonator.

Assembly Notes

I suggest mounting the lowest parts first (discrete resistors, speaker, axial capacitors, and diodes). Progress toward the taller parts (switches, radial capacitors, resonator, 0.3" wide ICs, etc.) Then the SIP resistors, transistor, and headers. Next, the socket strips for the 0.6" wide ICs, LEDs. Finally, the the battery holders.

Be sure pin 1 of all the ICs is in the lower left corner! Be sure to observe the polarity of the diodes, transistor, and tantalum capacitor C2! The SHORT lead of LEDs D4 and D5 goes in the hole closest to the 7-segment display.

There was no room for an expansion bus connector, but you can install a wire-wrap socket for U1. Its pins on the back can plug into a "daughter" board to provide access to all the signals needed for expansion. There are two mounting holes in the top corners to mount the mini H8, or attach an expansion board.

If you want to socket the 0.3" wide ICs, socket pins are the best option (Mill-Max #0552-1-15-01-11-27-10-0, digikey ED5037-ND, mouser 575-055210) for the very lowest height. Or if you use regular IC sockets, you'll have to use double sockets for the 0.6" wide ICs to have enough clearance.

This parts kit comes with a blank 27C256 32k EPROM.

You can use either a cheap CH340 USB adapter, or a Sparkfun FTDI adapter. The pinouts are DIFFERENT, so holes are provided for either one. Mount only ONE connector!

CH340: Use the white Molex KK connector. Cut off pin 3 (labeled 3V3 on the

adapter). Bend the remaining pins at right angles (away from the 2 "ears"), and install it with its holes parallel to the PCB so the adapter will lay flat on the board. The "ears" should be on top. You can cut them off if you like.

FTDI: Remove pin 2 from right-angle header. Install it with the pins pointing right, so the adapter can lay flat against the board.

There are two holes at the extreme right edge for a wire to go over the adapter to hold it down to the board.

3-pin header P3 sets whether there is ROM (normal) or RAM at address 0. Use a jumper to short the left two pins so the ROM will be at 0. The RAM position is ONLY used AFTER you have loaded a program into the (nonvolatile) RAM. (For example, CP/M which needs to have RAM at address 0).

3-pin headers P4 and P5 can be set to configure socket U3 for a 32k 27C256 or 64k 27C512 EPROM; or 128k 39SF010, 256k 39SF020, or 512k 39SF040 flash ROM. This parts kit comes with a blank 27C256 32k EPROM. Install it at U3 in the RIGHTMOST set of holes, so the left four holes are EMPTY. Jumper P5 to short the LEFT pair of pins. Jumper P4 to short the RIGHT pair of pins.

Testing

Connect a 5V supply with either a USB-serial adapter, or with an external 5V power supply. The pin header or Molex socket at P1 is a convenient place to connect power. The red POWER LED should light.

Turn switch S1 "ON". The board should draw about 25-30mA. If more, look for an IC in backwards or off by a pin, bad solder joints, etc.

Check the frequency at U8 pin 11 (RST7.5). It should be about 500 Hz (2 msec). The frequency is set by R2 and C5. These are low-precision parts: C5 is +/-20% and R2 is 5% so don't expect tremendous accuracy.

Check the frequency at U1 pin 37 (CLK0). It should be exactly 2.000 MHz. This proves the resonator and 80C85 clock is working.

Every location in the blank 27C256 EPROM is FFh, which is the 80C85 RST7 instruction. This instruction causes the 80C85 to continuously push the program counter into the memory locations pointed to by the stack register, and jump to address 0038h (which is of course ANOTHER RST7 instruction). Thus the 80C85 will be exercising RAM, ROM, /WR, and /RD. Check for pulses at U1 pins 31 (/WR) and 32 (/RD), and U5 pins 5 (RAM chip select) and pin 6 (ROM chip select).

If all is well, your mini H8 can run programs!

Theory of Operation

Y1 is a 4 MHz resonator with internal capacitors. The 80C85 has an internal inverter that forms a 4 MHz oscillator. This is divided by 2 to run programs. A 2 MHz clock is provided on U1 pin 37 (CLK0) if external circuits need it.

C1 and R1 provide a power-on Reset to U1 pin 37 (RIN). C1 is initially discharged (0 volts). When the SR or S0 keys are NOT pressed, R3B pulls the inputs of U8C low, so its output is high. R1 can then charge C1. After about 100 msec, the voltage on C1 is high enough to end the Reset and U1 starts executing programs. U1 pin 3 is an inverted version of this signal to U6 pin 35 (RESET).

There are two other ways to Reset U1. If power switch SW is switched to the OFF position, it grounds C1 to reset U1. Or, if BOTH the SR and S0 keys are pressed, both inputs of U8C go high, and the output of U8C goes low. This discharges C1 through R1 to cause a reset.

U2 is the low address latch. U1 (the 80C85) outputs the lower 8 address bits

on its AD0-7 pins in the first half of a bus cycle. U1 pin 30 (ALE) latches these bits so they are available in the second half of the bus cycle.

U5 is the address decoder. U5 pin 7 is enabled when U1 pin 34 (I/O/M) is "low" indicating that a memory cycle is occurring. U5 decodes address bits A13-A15 to decode eight 8k blocks (=64k). If a U5 input D0-D7 is "low", then pin 6 (W) will go "low" to chip-select U3 (ROM). If an input D0-D7 is "high", then pin 5 (Y) will go low to chip-select U4 (RAM). D0-D7 are thus wired high an low to put ROM at 0-8k (0-1FFF hex, 0-037.377 octal) and 32k-64k (8000-FFFF hex, 200.000-377.377 octal). RAM is at 8k-32k (2000-7FFF hex, 040.000-177.377 octal).

U5 pin 4 (D0) can actually be switched to put either RAM or ROM in the lowest 8k block with header P3 or U6 output bit PA4. But it is normally set "low" because you must have ROM in low memory to start up. This option is provided in case a program needs RAM at address 0 (like CP/M).

Once loaded, RAM contents are maintained by the batteries when SW is in the "off" position. Diode D2 lets the batteries ONLY power the RAM with SW "off". When VCC falls to 0v, everything in the circuit is at 0v. The positive end of the batteries will thus be 0v, so their negative end is at -4.5v, and RAM pin 14 (VSS) will be at -4.3v (due to the diode drop of D1). Thus the RAM is powered, and sees all its inputs "high" (i.e. the same as VCC). This holds its inputs (especially the important /WR) inactive.

Diode D1 only allows the cells to discharge, so you must use non-rechargeable cells. They also provide a higher voltage; the mini H8 would quit before rechargeables were actually dead.

R4 provides pullup resistors on /RD, /WR/ I/O/M, and RAM and ROM chip-selects because the 80C85 tri-states these lines during Reset.

The Heath H8 generated an Interrupt to restart the Monitor if BOTH the SM and S0 keys were pressed at once. Q1 and R3I provide this function on the mini H8. If S0 is pressed, the emitter of Q1 is high. If SM is pressed, it grounds the base of Q1. If both happen at once, Q1 turns on, its collector goes high, and generates an RST5.5 interrupt to U1.

Diode D3 allows the S0 key to also be read by U6 pin 12 (KMATH). Software can detect that S0 is pressed if KMATH is low and NONE of the 7-segment rows R1-R9 are low.

U8D, R2, and C5 provides the H8's 2 msec clock on U1 pin 7 (RST7.5 interrupt). U8D has Schmitt trigger inputs. The inputs must rise to about 2/3 of VCC for its output to go low, and must fall to about 1/3 of VCC for its output to go high. R2 and C5 cause the input to alternately charge and discharge to 1/3 and 2/3 VCC to produce a square wave output at about 500 Hz (2 msec per cycle).

The RST7.5 input of U1 is edge-triggered. Thus only the rising edge on U8D pin 11 will cause an interrupt. The interrupt can be cleared by resetting the RST7.5 bit with the SIM instruction.

U6 (82C55A) provides three 8-bit I/O ports to handle all the onboard I/O. U8B chip-selects it when A7=1 so it responds to any port address from 80-FF hex (200-377 octal). Use ports F0-F3 hex (360-363 octal) to match the Heath H8 Front Panel. Configure U6 in Mode 0 by writing Control Word 88 hex (210 octal) to port F3 hex (363 octal).

The 7-segment LED display is multiplexed almost exactly like the Heath H8. The lower 4 bits of output port F0 hex (360 octal) writes to U6 port PA0-3 to select the digits 0-9. U7 (74LS145) decodes the binary number 0-9 to set one of 10 rows R0-R9 "low". For example, write xxxx.0001 to set R1 low to enable digit 1.

Output port F1 hex (361 octal) writes to U6 port PB0-7 to select the segments. For example, light the A, B, and C segments to display a "7" by writing 00001110 binary, 0E hex (16 octal) to port F1 (361). The only

difference is that the mini H8 lights the segments with a "high", while the Heath H8 lit them with a "low". (I didn't have room for an octal inverter!)

The upper 4 bits of output port F0 (360) provide control functions. U6 pin 40 (Port A bit 4) selects the memory map. At Reset, PA4 defaults to an input, so jumper P3 determines whether address 0 is RAM or ROM. When U6 is initialized to mode 0, the port registers are reset to 0, so PA4 will be "low" to put ROM at address 0. It should always be set to 0 each time you write to port A.

U6 pin 39 (PA5) controls the "MON" LED. "High" lights the LED. Software should set it high when the monitor is running.

U6 pins 37-38 (PA6-PA7) control piezoelectric speaker SP1. It makes no sound when these bits are not changing. To make sound, set one bit high and the other low, and invert the two at the desired rate.

The keyboard is a lot different from the Heath H8. The keys are read by U6 pins 10-13 (bits PC4-PC7) on input port F2 (362). Keys SR and S0 can be read directly on PC6 and PC7 respectively. PC6 and PC7 are normally "low". When SR or S0 is pressed, the corresponding bit goes "high".

The rest of the keys are multiplexed with the LED display. When a row R1-R9 is selected, the corresponding key S1-S9 can be read on U6 pin 13 (PC4 bit 4, KNUM) on input port F2 (362). "High" means the key is not pressed; "low" means it is pressed. Similarly, the SP, S*, S-, S+, and SM keys are read on U6 pin 12 (PC5, KMATH) when rows R0-R4 are selected. Note that key SM is special as it is read with ANY row selected; so select row R0 to read it.

Serial I/O uses U1's SID and SOD pins. Serial input comes in on RX from a TTL 5v USB-serial adapter at P1. "High" is the "idle" state, and "low" indicates a Start bit or data. R4A holds RX "high" if no adapter is connected. U8A inverts the signal, so the idle state is low, and connects it to U1 pin 5 (SID) and interrupt pin 8 (RST6.5). The software must "big-bang" the serial data stream to decode the characters. The software can also use the RST6.5 interrupt to quickly detect a Start bit so as to not miss characters. Serial output data is also "bit-banged" and directly output on U1 pin 4 (SOD).

POWER LED D4 is simply wired to be "on" whenever there is 5v power. Resistor R5 sets its brightness.

That's it for now. Let me know what you think, what I missed, and anything that should be added! :-)

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